

**Autistic Postsecondary Students' Perceptions of Sensory Spaces in Academic
Libraries**

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As autism diagnoses rise, so do the numbers of autistic students enrolling in postsecondary education. This research will investigate the perceptions of autistic postsecondary students regarding sensory spaces in campus libraries, with a focus on gathering information about their opinions on inclusivity and support. Sensory spaces (also called multisensory or Snoezelen rooms/spaces/environments) are specially tailored environments in which users can stimulate or soothe their senses as needed. Their use is associated with positive behavioral changes in children and adolescents with sensory processing disorders. Because libraries are third spaces that are seen as welcoming to everyone and widely accommodating (Bertilsson Rosqvist et al., 2019), they present an ideal, less stigmatizing, venue to host a sensory space on a college or university campus.

While academic libraries are increasingly working to accommodate users with sensory differences by implementing new designs, equipment, programs, and services, few libraries currently contain dedicated sensory spaces. However, increasing numbers of academic libraries are creating dedicated sensory spaces that enhance or enable the library experience for autistic students. To date, little research has addressed autistic students' perceptions of these spaces, but as more academic libraries create them, closing this research gap is becoming critical. Sensory spaces must be tailored toward the specific needs and preferences of autistic students as expressed in their own words.

Autistic people experience the world differently than non-autistic people, especially when it comes to sensory input. Sensory differences can affect how autistic people learn, work, and socialize in various settings, including academic libraries. Although this proposal primarily focuses on sensory environments and their impact on autistic students, self-efficacy, which is an individual's belief in their own ability to succeed in specific situations, is inherently relevant to understanding how these students benefit from using sensory spaces. Individuals with higher self-efficacy have been shown to have better educational achievement and social relationships.

Preliminary research has found a possible link between an autistic adult's sensory environment and self-efficacy.

Creating sensory spaces in academic libraries is a laudable goal, but before these spaces can be established, more research must be undertaken. First, this study seeks to explore whether there is a relationship between the students' self-efficacy scores and the amount of time they spend using library sensory spaces. At this time, this researcher has not found evidence of a relationship in the literature, so it would be premature to investigate whether spending more time in sensory spaces improves an individual's self-efficacy or vice versa. The research hypothesis is:

- H1. There is a positive correlation between the self-reported self-efficacy scores of autistic postsecondary students and the amount of time these students use library sensory spaces.
- H0. There is no relationship between the self-reported self-efficacy scores of autistic postsecondary students and the amount of time these students use library sensory spaces.

In addition, to gain a holistic view of the sensory space experiences of autistic postsecondary students, this research will explore the following:

- RQ1. How do autistic postsecondary students describe the effects of utilizing sensory spaces on their self-efficacy in academic achievement and social interactions?
- RQ2. How do autistic postsecondary students' sensory needs and preferences align or fail to align with the design and offerings of sensory spaces?
- RQ3. What suggestions and feedback do autistic postsecondary students have for improving sensory spaces in academic libraries?

Through the first-person perspectives of autistic students, this mixed methods study aims to contribute to the literature on sensory spaces and academic libraries, as well as to the empowerment and self-advocacy of autistic students. Centering the voices of autistic postsecondary students will illuminate how sensory spaces meet the students' needs, barriers to utilization, and the potential for utilization of these spaces to enhance academic engagement and social well-being.

Literature Review

Autism, officially known as Autism Spectrum Disorder (ASD) is a complex developmental condition associated with characteristics such as communication deficits, dependence on routines, repetitive behaviors, restricted interests, resistance to change, and sensory sensitivities (American Psychiatric Association [APA], 2022; Van Hees et al., 2015). ASD is called a "spectrum" disorder because there is wide variation in the type and severity of symptoms people experience (Everhart & Escobar, 2018). According to the Centers for Disease Control and Prevention (CDC), as of 2020, about one in 36 children was diagnosed with ASD, up from one in 44 in 2018 (2023). As the diagnosis rate continues to rise, so too do the numbers of autistic students embarking on postsecondary education journeys (Anderson, 2018; Anderson et al., 2018). Attending college or university can be beneficial for autistic adults, as those who graduate earn more money and experience a better quality of life than their peers who did not graduate (Anderson et al., 2018). However, these students face many challenges, contributing to lower enrollment rates compared with their peers, including disabled students without intellectual impairments (Anderson et al., 2018), and lower graduation rates compared with their allistic peers (Anderson, 2021).

The transition to postsecondary education presents numerous challenges for autistic students, as they encounter many barriers due to their struggles with executive function and social and communication deficits (Anderson et al., 2018; Cho, 2018; Shattuck et al., 2014; Shattuck et al., 2017; Sturm & Kasari, 2019). Most studies of autistic students were conducted with K-12 students in the education field. (Anderson, 2018; Braumberger, 2021; Cho, 2018). However, there have also been several studies that identified challenges faced by autistic postsecondary students. The scope of this literature review is limited to library science papers that identified challenges and supports to autistic postsecondary students. In terms of executive functioning, these students encounter difficulties with emotional regulation, self-regulation, and self-efficacy (Anderson, 2021; Cho, 2018; Shattuck et al., 2014). They have a tendency to miss the big picture due to getting bogged down in details, and succumb to overly rigid thinking (Braumberger, 2021; Cho, 2018). In addition, sensory sensitivities can make it

difficult to concentrate or focus on anything except the source of discomfort (Anderson et al., 2018; Cho, 2018). In terms of communication and social skills deficits, a meta-analysis by Gelbar et al. (2014) found that autistic college and university students identify their social skills deficits and high levels of anxiety, loneliness, and depression, as their most pressing challenges. They also struggle with self-efficacy, which includes tasks such as knowing when and how to advocate for themselves (Anderson et al., 2018).

These challenges are even greater for those with other complicating factors. Comorbid forms of neurodiversity, including ADHD and dyslexia, are common in autistic people, as are mental health and behavioral struggles, especially anxiety (APA, 2022; Anderson et al., 2018; Gelbar et al., 2014). Female autistic students represent another population that struggles significantly more. They experience more bullying, anxiety, and depression, and have reported lower levels of satisfaction with their universities' autism support and university life (Anderson et al., 2018; Sturm & Kasari, 2019). Similarly, autistic youth of color are diagnosed and begin treatment at older ages than their white peers (Pinder-Amaker, 2014) and have poorer educational outcomes (Riley, 2010). Research into the unique needs of individuals in these populations is critical.

Many people with ASD process sensory input in an atypical manner. Sources of routine, background stimulation, such as the sound of a fan or a damp handshake, may be felt more intensely or be more difficult to filter out. An autistic person might hyperfocus on specific sensory details, or be under-responsive to sensory stimuli (Remy & Seaman, 2014). To cope with these challenges, autistic people often engage in stimming, which is a repetitive, self-stimulatory behavior that can help manage sensory overload. Unfortunately, stimming behaviors are often viewed as socially unacceptable to do in public, and many autistic individuals therefore "mask" or hide their behavior by reducing, eliminating, or substituting their preferred stims. Masking increases cognitive load and reduces self-efficacy (Nwaordu & Charlton, 2023).

Sensory spaces are specialized areas designed to stimulate or soothe a person's senses, usually through special lighting, music, and objects. They can be used therapeutically for children and adults with autism, sensory processing disorders, or

other developmental disabilities. Other individuals who may benefit from using a sensory room include people with ADHD, anxiety, or dementia. Originating in the Netherlands in the 1970s, sensory spaces were originally designed as a safe leisure activity for profoundly physically or mentally disabled individuals. (Botts et al., 2008). Typical sensory room equipment includes visual stimuli such as dimmable or color-changing lighting, projectors, fiber optics, and LED lights; auditory stimuli such as sound machines and noise-cancelling headphones; and tactile stimuli such as textured walls, floors, and weighted blankets. Autistic individuals can also privately engage in their preferred stimming behaviors, as long as they are not physically harmful in sensory spaces. Although this researcher has found few studies that investigated the outcomes or impacts of utilizing sensory spaces, preliminary results suggest that autistic children engage in fewer disruptive behaviors, such as stimming, after spending time in a sensory environment (Kim & Park, 2021). No research addresses how autistic adults of average or above average intelligence experience sensory spaces in their own words.

About half of all autistic people have average to above-average intelligence (Anderson, 2021). Autistic people are significantly more likely to major in STEM fields than the general population or students with other disabilities (Shattuck et al., 2014). They have a strong ability to focus, attention to detail, good memories, dedication, perseverance, analytical skills, observational skills, passionate interests, a desire to find the right answers, enthusiasm for following rules, and an affinity for technology (Anderson et al., 2018; Anderson, 2021; Van Hees et al., 2015). These strengths mean that autistic students might struggle with academics in different ways than their allistic peers (Anderson et al., 2018).

As more autistic students have pursued postsecondary education opportunities, colleges and universities have recognized the need to adapt their services and spaces to better serve these students. Suggestions for improving outcomes include the development of support programs that benefit autistic postsecondary students (Anderson, 2021; Shea & Derry, 2022). In developing these support programs, institutions have drawn on insights from the education field showing that “autistic students benefit from multi-sensory teaching styles, increased visual content, more

written explanations, simplified processes, reflective activities in place of social ones, linking new information to familiar information, explanations of classification systems for locating materials, and individualized appointments” (Remy & Seaman, 2014). Common services available through autism support programs include academic advisement and tutoring, counseling, peer mentoring; support curricula that provide time management, social, organizational, and self-advocacy skills. Less common services include life skills education in areas such as financial management, chores, community education, and transportation (Anderson et al., 2018; Anderson et al., 2019).

Often, these students require just as much social and emotional support as academic support (Anderson, 2021; Gelbar et al, 2015; Van Hees et al., 2015). The institutions they attend have adopted the language of acceptance and inclusion, but there continues to be a mismatch between the emphasis on academics by most of autism support programs and the need for non-academic supports in these programs (Anderson, 2018; Anderson et al., 2018). Unfortunately, the non-academic supports that exist (not including sensory spaces) have been rated less helpful than the academic supports (Anderson et al., 2018), and undiagnosed and non-disclosing students are more difficult to support because they cannot easily be identified (Anderson, 2018).

Anderson et al. (2018) extensively surveyed 40 autistic university students about their experience of university-provided support. However, the researchers failed to ask these students about any physical or sensory adaptations or supports they might need or currently take advantage of, despite half of their respondents mentioning sensory sensitivities that interfere with their lives. The omission of questions about sensory support may be at least partially because sensory sensitivities were not described as a characteristic of autism spectrum disorder (ASD) until 2013 when the DSM-5 was issued. A need for more research into how autistic postsecondary students feel about non-academic supports, such as sensory spaces, is indicated (Braumberger, 2021; Cho, 2018; Everhart & Anderson, 2020; Pionke et al., 2019). Are colleges and universities failing to provide necessary physical accommodations, or are autistic students failing to use them, and what are their reasons?

Autistic postsecondary students value the library as a safe and welcoming place, appreciating the library's resources, services, and programs that support their academic and personal interests (Anderson, 2018; Shea & Derry, 2019). They expect to be able to find quiet spaces in libraries to escape sensory overload (Anderson, 2021); one student called his school's library his "sanctuary" (Anderson, 2018, p. 645). As a result, libraries are well-positioned to create more inclusive and accessible library spaces, services, and programs for autistic students (Anderson, 2021). The most frequently reported benefits of providing academic library services for autistic students were increased inclusion, diversity, and accessibility, as well as improved student satisfaction and retention (Shea & Derry, 2022). Libraries are open to all, and a good, neutral place for semi-structured social interactions such as special interest club and study group meetings. Semi-structured socializing can be easier for autistic individuals because there are explicit rules to follow (Braumberger, 2021). Because autistic people are often familiar with libraries and less amenable to new experiences, the on-campus library can be a bridge that is just familiar enough for autistic students, increasing the chances that they will visit a space that is oriented toward their needs.

However, the accessibility of academic libraries for autistic students could be improved. Van Hees et al. (2015) pointed out that autistic students may be sensitive to common library occurrences like flickering lights, clacking keyboards, and random movements. Poor lighting, noise, confusing signage, and other sensorily-taxing elements of the built environment can lead to sensory overload as well (Shea & Derry, 2019). Yet spaces that are too quiet and isolated can also contribute to feelings of loneliness and depression (Anderson et al., 2018; Gelbar et al., 2015). Some autistic students also report running afoul of unspoken social rules in the library, for example, speaking too loudly in a quiet area. Misunderstandings caused by librarians not being familiar with autism as an invisible disability have also resulted in difficult social situations (Braumberger, 2021). Only about a third of academic libraries have any kind of support program for autistic students. Common academic library services for autistic students include quiet study spaces, noise-canceling headphones, sensory kits, and social stories (Shea & Derry, 2022).

Braumberger notes that “while the literature is relatively unanimous in acknowledging that sensory issues are a concern for autistic students, the observations and recommendations for improvement are from limited data sets” (2021, p. 97). Only a few studies included information about accommodations specifically designed for sensory sensitivities. Anderson’s (2018) content analysis of posts by users on the autism-focused website WrongPlanet revealed that autistic postsecondary students are sensitive to the noise or lack thereof in libraries. Cho (2018) wrote about The Bridges to Adelphi program at Adelphi University, a comprehensive autism-support program. This program established a combination sensory room and social space in the basement of a dorm. However, while Cho’s overview addressed faculty and staff experiences with the program, the perspectives of the students themselves were not covered. A literature review found that spaces tailored for autistic students must be (1) single-purpose usage (2) provide consistency and predictability (3) have low levels of sensory stimulation (Walton & McMullin, 2021).

Anderson (2021) found that when libraries are involved in support programs for autistic students, they must be critical collaborators for students to benefit. Partnerships ensure that autistic students’ needs are holistically addressed (Braumberger, 2021). Other on-campus organizations must be aware that librarians are willing, able, and have a lot to offer autistic students beyond physical space. That said, “providing access to physical space was the main role that libraries and librarians provided for students” (p. 108, Anderson, 2021). Libraries can use the physical space as a jumping off point or central hub for a campus-wide autism support program. However, not enough attention has been paid to designing the physical environment to best accommodate a range of students, including autistic ones. Since 2015, more academic libraries have adapted their spaces so that the design is friendlier to those with sensory sensitivities. Sensory rooms can also help alleviate anxiety and social anxiety by being spaces where users are free from observation by others (Braumberger, 2021).

Most studies of postsecondary autism supports have been qualitative studies of subjects in the US, the UK, or Australia, with small sample sizes. Many lack the perspectives of students and librarians with autism, relying instead on information

gleaned from librarians, educators, or other individuals who work with autistic students (Anderson et al., 2018; Anderson et al., 2019; Braumberger, 2021). No other study uncovered during the literature review focuses specifically on sensory spaces, which have emerged in academic libraries as important resources aimed at addressing the unique needs of autistic postsecondary students. Despite their growing prevalence, there remains a significant gap in the literature regarding the subjective experiences of how and why students utilize these spaces. Understanding these experiences is essential to providing insight into the effectiveness, accessibility, and impact of these spaces from the perspectives of the individuals they are designed to support.

Theoretical Framework

The theoretical foundations of this study are rooted in Bandura's social-cognitive theory, which posits that self-efficacy plays a pivotal role in influencing a person's resilience; the social model of disability, which represents a paradigm shift from traditional views of disability; and Universal Design for Learning (UDL), which complements social-cognitive models by providing a framework for creating environments that accommodate individuals' varied learning styles.

Social Cognitive Theory

To date, social-cognitive theories have provided the primary frameworks for understanding and treating ASD because ASD is diagnosed by observing an individual's behavior; there is no genetic test for autism (Livingston & Happé, 2021). In particular, Bandura's social-cognitive theory, which emphasizes the relationships between individuals' self-efficacy and their environment, offers a valuable lens for examining how autistic students perceive and utilize sensory spaces in libraries. The social-cognitive theory, with its focus on the role of observational learning, social interactions, and cognitive processes, underscores the importance of creating environments that allow students to increase their belief in their own capabilities (Feldhaus et al., 2015).

Strengths of this theory include the model's holistic approach to analyzing the interactions between an individual's behavior and their environment and its emphasis on self-efficacy, which has been shown to influence motivation and behavior. Critics

charge that social-cognitive theories are insufficient because they do not account for biological factors.

Social Model of Disability

Historically, disability was viewed through a medical lens, focusing on individual limitations and advocating for treatment or rehabilitation as the primary means of integration into society. This perspective, known as the medical model of disability, inherently positions disability as a problem to be fixed. In contrast, the social model, emerging prominently in the 1970s and gaining traction through disability rights movements, posits that disability is not caused by individual impairments but by societal barriers and attitudes that fail to accommodate diversity in human functioning. This model emphasizes environmental, structural, and attitudinal barriers as the main factors contributing to disability, advocating for societal changes to remove these barriers (Anderson, 2018; Braumberger, 2021).

The strengths of the social model lie in its empowerment of people with disabilities, promoting a sense of agency and advocacy for systemic change. According to the social model of disability studies, self-advocacy is critical to empowering differently able individuals to meaningfully participate in increasing inclusivity. Accordingly, this paper makes use of identity-first language ("autistic person") instead of person-first language ("person with autism"), which emphasizes that autism is a core part of an autistic person's identity, and there is nothing wrong with that. Identity-first language is also the preference of many autistic people (Braumberger, 2021). Several researchers, including Anderson (2018) and Shea and Derry (2022), have used the social model of disability to inform their study designs, performing qualitative or mixed methods research that centers the voices of autistic postsecondary students.

The social model has been instrumental in fostering disability rights movements and shaping inclusive policies and practices. However, the model also faces criticism for potentially overlooking the actual experience of living with disabilities, including personal and medical aspects of living with impairments. Critics argue that it may underplay the importance of medical support and the personal challenges associated with disability. Furthermore, the model's strong emphasis on societal change, while

crucial, can sometimes neglect the immediate and practical needs of individuals with disabilities.

Universal Design for Learning

Originating from the concept of universal design in architecture, Universal Design for Learning (UDL) applies the same principles to the field of education. Just as the built environment should be designed to be widely accommodating, so too should curricula be developed to accommodate a wide variety of learning styles. UDL's strengths lie in its proactive approach to designing educational experiences that anticipate the needs of a wide variety of students, including those with disabilities, thus reducing or eliminating the need for ad-hoc accommodations or modifications (Anderson et al., 2018; Cho, 2018; Pionke, 2017). UDL is an especially important guiding philosophy because not all students with autism or disabilities disclose their diagnoses (Braumberger, 2021), but universal design means that the space and services are planned to be inclusive of their needs as well. Because not disclosing is fairly common, researchers promote UDL as a way to frame their suggestions for making physical spaces and library services more inclusive that does not require making special adjustments based on individual needs, but instead serves the widest number of students possible in its default state (Pionke, 2017; Remy & Seaman, 2014; Shea & Derry, 2019). Criticisms of UDL focus on the fact that it is virtually impossible to accommodate the needs and preferences of all learners.

The integration of the social model of disability with UDL offers a robust theoretical basis for examining the experiences of autistic postsecondary students in sensory spaces in academic libraries. The social model provides a lens through which to view societal and environmental barriers, while UDL guides an examination of how effectively sensory spaces meet the sensory needs of their users and emphasizes the importance of proactively identifying and eliminating social and environmental barriers.

Methodology

This will be a mixed methods study, combining quantitative and qualitative analysis. Mixed methodology enables a more comprehensive exploration of autistic students' experiences and perceptions. While quantitative data provides measurable evidence of the prevalence of issues with self-efficacy that can be generalized to the

broader population of autistic postsecondary students, qualitative data provides deeper insights into students' experiences, attitudes, feelings, and perspectives. This dual approach ensures a holistic understanding that either method alone might miss.

Self-efficacy, a measure of an individual's coping skills, is typically determined through the use of quantitative questionnaires (Feldhaus et al., 2015; Nwaordu & Charlton, 2023; Shattuck et al., 2014). Using a questionnaire allows for a standardized measure that can be generalized to a wider population, but because questionnaires are self-reporting measures, subjectivity and bias may creep into the results (Feldhaus et al., 2015).

Qualitative research, which is distinguished by its focus on understanding human behavior, experiences, and motivations, is particularly valuable for exploratory research fields where prior research is scarce. This approach emphasizes meaning-making, offering rich, detailed insights that are unattainable through quantitative methods alone. In the context of studying the under-researched experiences of autistic postsecondary students with sensory spaces in academic libraries, qualitative methods will enable the exploration of complex, subjective experiences from the perspectives of those directly impacted. Students will, in their own words, describe how their sensory needs and preferences are met through the use of a sensory space and give their suggestions for improvement.

The researcher made this methodological choice for several reasons. First, qualitative research emphasizes the perspectives of participants, making it uniquely well-suited for exploring the subjective experiences of autistic students, whose sensory perceptions and needs may differ significantly from the allistic population (Bruamberger, 2021). Second, the flexibility of qualitative research allows for the exploration of complex phenomena within their natural settings, enabling a comprehensive understanding of how sensory spaces in libraries impact autistic students' ability to study and feel comfortable within these environments (Pionke et al., 2019). Finally, including the direct perspectives of autistic students will result in more comprehensive and meaningful insights (Bertilsson Rosqvist et al., 2019). This approach aligns with the study's aim to identify sensory elements that are most

beneficial or challenging, as it provides the depth and context necessary for understanding these elements

This study will use a phenomenological approach to increase understanding of how autistic postsecondary students perceive and utilize sensory spaces in academic libraries. Using mixed methods reflects a broad commitment to research that is deeply informed by the perspectives of those it seeks to understand, thereby contributing to the literature and to the improvement of inclusive practices.

Positionality Statement

In approaching a study about the perceptions of autistic college students, it is important to consider how personal identities and experiences have shaped the study's research questions and methodology.

The confluence of being poor, Black, queer, and having ADHD in a society that marginalizes those identities provides me with a distinct lens through which I view accessibility and inclusivity, guiding me toward a more inclusive, participatory, and intersectional research design. Although I was interested in neurodiversity and diversity in learning styles long before I met him, being close with my fiancé also gave me the opportunity to closely observe the struggles and unique strengths that were associated with his autism. His transition to college, like mine, was difficult. Unfortunately, he found even less support than I did; he did not graduate and was underemployed all of his adult life, a common outcome for unsupported autistic students. Although I am not a part of the autistic community, I do have some understanding of small parts of the struggles faced by autistic individuals as a member of the wider community of non-neurotypical individuals by dint of my ADHD diagnosis and seeing the struggles of my fiancé.

My background has prepared me to see how much academic research, especially in autism studies, has focused on how to interact with white male subjects. To remedy that, it is essential to hear directly from people with autism and to introduce greater diversity to subject pools to reflect the diversity of the people who actually live with autism. Though I do not expect much age range diversity in a study focusing on

undergraduates, there should be diversity in the genders, races, socioeconomic and religious backgrounds, and sexual orientations of the participants.

My struggle with transitioning to college due to inadequate support structures is a shared reality for many who find themselves at the intersection of multiple marginalized identities as well. My experiences have allowed me to see how existing support systems in academic library environments are often designed without considering the populations they are designed to support. Sensory spaces may not be fully accessible to students who navigate the world with multiple marginalized identities if they are not designed using an intersectional approach that seeks to understand social and sensory issues from their perspectives.

Ethical Considerations

This research is being conducted with a vulnerable population, making it particularly important to carefully plan a transparent and ethical project. The study involves human subjects and will require an expedited IRB review, since it poses no more than a minimal risk to the subjects, involves collection of non-anonymized data from image and video recordings, and researches behavioral characteristics. Confidentiality and anonymity are of paramount importance, as an autism diagnosis is private medical information. Identifying information will be stored and backed up in secure electronic form, and access to that data will be limited to the primary researcher. Given the use of Photovoice, which involves photography that may include identifiable locations or indirectly reveal participants' identities, photographic images will not be released; only anonymized interview transcripts will be made accessible to researchers other than the primary investigator.

Because autistic individuals are more prone to anxiety and depression, special care must be taken to ensure that each subject is and feels supported. For example, every study facilitator will receive training on best practices on communicating with autistic people, and information on autism and mental health resources will be distributed at the focus group. The information on communication and health resources will also be posted on the study website so participants can review it on their own time in a comfortable setting. The focus groups will take place in a low-stimulation sensory

environment with dim lighting, comfortable seating, and few distractions. Each participant will be given a sensory kit they can use for self-soothing during the discussion. Participants will be assured throughout each session that they are free to leave at any time.

Cultural sensitivity and familiarity with autism and autistic people are critical attributes for the researchers conducting the Photovoice focus groups. Cultural sensitivity is vital for building trust, ensuring respectful and meaningful engagement with participants, and accurately interpreting data.

Because people have different learning styles, informed consent and all information about the project will be disseminated in various formats. Large print plain language information, a video with the same information conveyed orally and visually, and step-by-step pictures, will be offered in person and also be accessible at any time on the project's website. Informed consent procedures will clearly explain how the images will be used and stored, and that they will not be shared to safeguard participants' anonymity and privacy. It will be explained that participation in this project is voluntary, and anyone may withdraw at any time for any reason with no negative consequences.

Prior to publication, feedback about the study's findings will be made available to participants via the study's website. Through the website, participants will also be informed of any real-world changes made to the university libraries' sensory spaces that were made because of the insights gleaned from this study.

Sampling and Recruitment Strategies

This study sample will consist of diagnosed and undiagnosed autistic postsecondary students at a large southeastern university. This selection was made in order to gather information directly from the perspectives of the population who will make use of sensory spaces. Although it is impossible to know the exact size of the autistic student population, this study will use the current CDC estimate of the percentage of autistic adults to estimate the number of autistic students at the university. The target population size is therefore estimated to be 2.21% of 36,000 students at the university, or 796 students (CDC, 2022).

The researchers will use a combination of snowball sampling and stratified sampling to find as many autistic students as possible across a wide demographic range. Snowball sampling involves having existing study participants recruit future subjects from among their network (Connaway, 2016). Researchers will begin by using flyers, emails, posters, and working with the on-campus Office of Disability Support (ODS) and Autism Pride group to recruit an initial group of officially diagnosed and self-diagnosed participants. This group will be incentivized via five-dollar gift certificates to recruit study participants, who will themselves receive a \$25 gift certificate for participating. Every student who recruits another will earn an additional \$5 per qualified participant. This method will help the researchers reach autistic students who do not interact with the ODS, especially self-diagnosed students.

To ensure the sample is representative across demographics, snowball sampling will be combined with stratified sampling. The student population will be stratified based on variables such as their year of study, major, and frequency of library usage. After initiating the snowball sampling process, the demographics of participants will be carefully tracked, and efforts will be made to find students from underrepresented groups. The study will aim for a representative sample of at least 50% of the target population, or 398 students.

Data Collection Plan

This study will use a self-efficacy survey instrument and a Photovoice focus group to collect quantitative and qualitative data. Quantitatively measuring self-efficacy is typically done using self-reported questionnaires. A well-known instrument for this purpose is the General Self-Efficacy Scale (GSES), developed by Schwarzer and Jerusalem (1995). This scale assesses a broad and stable sense of personal competence to deal effectively with a variety of stressful situations. Items on the GSES are designed to tap into the core elements of Bandura's social-cognitive framework, specifically the belief in one's own ability to respond to new or difficult situations and to deal with associated setbacks. The GSES has been found to be a robust measure, and studies measuring the self-efficacy of autistic adults have garnered reliable results (Feldhaus et al., 2015; Nwaordu & Charlton, 2023; Shattuck et al., 2014).

Photovoice, a qualitative research strategy developed by Caroline Wang and Mary Ann Burris (1997), empowers individuals to document and share their own experiences through photography followed by narrative explanation, thereby capturing aspects of their lives and environments that are meaningful to them. Photovoice facilitates a deeper understanding of personal experiences, literally making overlooked or misunderstood aspects of participants' lives visible. Previous studies employing Photovoice with autistic participants have successfully highlighted personal narratives and experiences, helping autistic participants overcome communication barriers to provide valuable insights into their daily lives, challenges, and the environments they navigate (Do et al., 2021; Krutt et al., 2018; Lam et al., 2020). Using a similar technique, Everhart and Escobar (2018) gave students Go-Pro cameras and had them talk through their decision-making while completing tasks in the library.

In this study, the first step will be to recruit and train facilitators. Facilitators must be familiar with ASD and will be trained in best practices on interacting with autistic individuals. After recruitment, participants will be emailed a survey instrument with demographic questions and a self-efficacy questionnaire. After completing the survey, participants will attend an introductory workshop at which the Photovoice process will be explained. Participants will be given guidelines on what types of images to capture to illustrate their experiences with sensory spaces. Participants, who will receive tips on photographic composition, will use their own phones to take photographs.

Next, participants will take photographs over a two-week period, focusing on depicting academic and social successes and challenges, and identifying elements that impact their sensory experiences in academic libraries. This might include areas that are particularly comforting or challenging.

After the photographing phase, each participant will meet individually with a facilitator over Microsoft Teams or Zoom. Together, they will select 10 to 15 photographs that are most representative of the student's experiences, and the facilitator will help the student prepare a brief narrative explaining why the image was chosen and what it represents. Next, students will meet in focus groups of two to four

participants, plus a facilitator, to discuss the photographs and their narratives in a structured focus group. This collaborative environment facilitates the discovery of shared experiences and perceptions, which helps to create a deeper understanding of the sensory challenges and preferences of autistic postsecondary students in academic library sensory spaces.

The focus group questionnaire was developed based on principles originally outlined in Wang and Burris (1997) and refined by Sutton-Brown (2014). Incorporating best practices in Photovoice methodology, the questionnaire emphasizes participant-led inquiry and reflexivity to prompt students to explore their sensory experiences, needs, and suggestions for improvement. The questionnaire guides discussions, stimulates critical thinking, and encourages detailed feedback.

Data Analysis Plan

This study will generate the following raw data:

1. General Self-Efficacy Scale (GSES) survey results (quantitative)
2. Demographic data (quantitative)
3. Photographs taken by the study participants (qualitative)
4. Narrative explanations of the photographs by the study participants (qualitative)
5. Audiovisual recordings of focus group discussions (qualitative)

This study will employ a mixed-methods approach to data analysis. The participants' belief in their ability to achieve academic and social goals will be quantitatively measured by the adapted General Self-Efficacy Scale (GSES). The GSES is a well-established survey instrument that quantifies a person's belief in their ability to achieve goals. The GSES was adapted by Shattuck (2014) to investigate the students' self-efficacy as it pertains to their academic and social skills, and some of those questions are utilized in this study. Descriptive statistics will be utilized to obtain a general understanding of the participants' self-efficacy levels. In addition, inferential statistical tests will be conducted to uncover deeper insights. Pearson's correlation coefficient will be used to measure the strength and direction of the relationship between self-efficacy scores and students' utilization of sensory spaces.

In addition to completing the demographic and GSES questionnaires, the study participants will generate a variety of qualitative data: photographs taken by study participants, narrative explanations of the photographs, and transcriptions of audiovisual recordings of focus group discussions. This qualitative data will be analyzed using various machine learning techniques.

The focus group transcriptions will be cleaned and standardized in preparation for analysis. This corpus will then be analyzed using Natural Language Processing (NLP) techniques. The text will be coded using a supervised machine learning model to identify themes. First, human reviewers will develop a training data set by manually coding a portion of the data. This manually coded data will serve as the basis for training a classification model, which will learn to code the rest of the data automatically. The model will be created using Python. To ensure the validity of the analysis, 10% of the computer-coded data will be randomly selected to be independently re-coded by human reviewers, and the results will be compared to the machine-generated data.

Using machine learning for content analysis will enable the researchers to more efficiently analyze the vast amount of data that the participants will generate throughout the course of this project, providing nuanced insights into the sensory experiences of autistic students.

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Appendix A
Demographic Questionnaire

Age

Gender

Sexual Orientation

Major

Academic Level (Freshman, Sophomore, Junior, Senior, Graduate Student, Professional Student)

Ethnicity

Race

Residential Status (on or off campus)

Age of Diagnosis

Type of diagnoses (self-diagnosed or professionally diagnosed)

Library usage (hours in the last week)

ODS usage (hours in the last week)

Sensory space usage (hours in the last week)

General Self-Efficacy Questionnaire

1 = Not at all true 2 = Hardly true 3 = Moderately true 4 = Exactly true

1. I can always manage to solve difficult problems if I try hard enough.
2. If someone opposes me, I can find the means and ways to get what I want.
3. It is easy for me to stick to my aims and accomplish my goals.
4. I am confident that I could deal efficiently with unexpected events.
5. Thanks to my resourcefulness, I know how to handle unforeseen situations.
6. I can solve most problems if I invest the necessary effort.
7. I can remain calm when facing difficulties because I can rely on my coping abilities.
8. When I am confronted with a problem, I can usually find several solutions.
9. If I am in trouble, I can usually think of a solution.
10. I can usually handle whatever comes my way.
11. I can get school staff and other adults to listen to me.
12. I know how to get information I need.

Focus Group Questionnaire

1. Introduction

- a. Distribute materials with an explanation of the study, the address of the study's website, mental health and autism support resources, and statement of confidentiality.
- b. Explanation of the study: This research project will investigate how current sensory spaces within academic libraries meet or fail to meet the sensory needs and preferences of autistic postsecondary students by collecting detailed feedback from these students. A secondary goal of the project is to empower autistic students to use their voices to advocate for changes that make sensory spaces more inclusive and supportive.

This study uses Photovoice to collect data. Photovoice is a form of research that uses participant-created photography to help participants to describe their environment and experiences. This method is especially useful in engaging marginalized groups in decision-making processes about issues that impact their lives.

Through group sessions like this, study participants will discuss their photos, explaining what the images represent and sharing personal narratives. The final aim is to use the insights gained through these discussions to advocate for changes that will make sensory spaces more accommodating and beneficial.
- c. Explain to participants that their photographs and the discussions are private and confidential. Explain that their discussions will be anonymized before analysis. Explain that they have the right to withdraw from the study at any time. Ask if there are any questions.

2. Part 1: Sensory Spaces and Self-Efficacy

- a. Please share a photo that represents an achievement or accomplishment. What does this photograph reveal about your experiences?
- b. Please show us a photo that depicts a challenge or obstacle you have faced. Did you overcome the situation? How did you handle it?

- c. Can you share and describe for us a photo that represents the support you receive at university? What are some positive aspects of this support? Drawbacks?
 - d. Can you provide a photo that illustrates a time when you had to advocate for yourself or others? What did you learn from this experience?
- 3. Part 2: Understanding Sensory Needs and Preferences
 - a. Please share a photo that represents an aspect of a sensory space that you find helpful. What about this aspect works well for you?
 - b. Please share a photo of a place that is challenging for you due to your sensory needs. What makes this space challenging?
 - c. Can you share a photograph that illustrates how you interact with sensory spaces in the library? What story does this photograph tell?
 - d. Reflecting on your sensory experiences, are there spaces within the library that challenge your sensory preferences? Please share examples through your photographs.
 - e. How do these spaces impact your study habits, well-being, and academic performance?
- 4. Part 3: Exploring the Design and Offerings of Sensory Spaces
 - a. Using your photographs as a reference, please identify design elements or offerings in the library's sensory spaces that are beneficial to you.
 - b. Using your photographs as a reference, please identify design elements or offerings in the library's sensory spaces that are problematic for you.
 - c. Can you describe what inclusivity and accessibility mean to you? What do your photographs reveal about the inclusivity and accessibility of the library's sensory spaces?
 - d. Discuss any gaps or missing elements in the sensory spaces that your photographs have highlighted.
- 5. Part 4: Suggestions for Improvement

- a. Looking at the issues identified through your photographs, what changes or improvements would you suggest for sensory spaces in the library? Can you identify specific design features or resources?
 - b. How do you think these improvements might impact your academic and social experiences within the library and on campus as a whole?
- 6. Conclusion
 - a. Thank the participants for their contributions.
 - b. Reiterate that the participants' discussions and photographs are confidential.